Vishal Sharma

Contact Information

525 Lexington Avenue, New York, 10017
+1-435-764-5663
DOB: Oct 13 1988, Citizen: India

Email: vishalsharma16@acm.org
Memberships: ACM, IEEE Senior
•Website • Profile • Github • Kaggle • Quora

Education & Training

Postdoctoral Research Fellow - Harvard Ophthalmology AI Lab — 2021 - 2022
Harvard Medical School/MEEI, Massachusetts, USA
Mentor: Mengyu Wang, PhD (Assistant Professor, Harvard Medical School/MEEI)

Doctor of Philosophy, **Computer Science** — 2016 - 2021 | 4.0
Utah State University, Utah, USA
Dissertation: Deep Learning Data and Indexes in a Database (over 1,500 downloads)
Mentor: Curtis Dyreson, PhD (Professor)

Master of Science, **Computer Science** — 2012 - 2014 | 3.91
Utah State University, Utah, USA
Thesis: MultiverseJava<temporal>: Programming Databases with Interesting Values
Mentor: Curtis Dyreson, PhD (Professor)

Bachelors of Technology, **Computer Science and Engineering** — 2006 - 2010 | 8.24
SRM Institute of Science and Technology, India (ranked 13 by the Ministry of Education, 2024)
First Class with Distinction (formerly SRM University)

Academic Appointment

Asst. Prof. TT, Big Data Science - University of Nevada, Las Vegas — Feb 2024 - July 2026
- Part of campus-wide cluster hire. Developed four courses on Data Science for Healthcare track, part of a certificate program for Healthcare Information Technology
- Teaching: (i) Data Mining for Healthcare (rating 4.67/5) (ii) AI for Healthcare (4.35/5)

Asst. Prof. TT, Computer Science - New York Institute of Technology (NYIT) — Sept 2023 - Jan 2024
- Teaching: (i) Introduction to Data Mining (rating 4.8/5) (ii) Database Systems (rating 4.5/5)

Research Associate/Scientist - Cincinnati Children's — Oct 2022 - Dec 2022

Teaching Experience

2025	Instructor (4.67/5)	Ph.D. & Masters	Data Mining for Healthcare
2025	Instructor (4.35/5)	Ph.D. & Masters	Artificial Intelligence for Healthcare
2024	Instructor (4.41/5)	Ph.D. & Masters	Data Mining for Healthcare
2023	Instructor (4.5/5)	Ph.D. & Masters	Database Systems
2023	Instructor (4.8/5)	Masters & UG	Introduction to Data Mininig

Grants Preparation

NIH R01 - Quantifying the impacts of social media on global mental health (under preparation)
NSF CAREER - Framework to Support The Process of Discovery (under preparation)
NSF ART - Virtual Reality (VR) Accelerated Ophthalmology (under preparation)

Professional Experience

Co Founder & CTO - Vaision, USA — 2023 - 2024
- Vaision has been concluded

Research Intern - Intel, USA — Summer 2019
- *Mentor*: Don Kent (Senior Manager Data Science)

Data Scientist Intern - IM Flash (Intel-Micron), USA — Summer 2018
- *Mentor*: Pradeep Ramachandran (Senior Member of Technical Staff)

Senior Software Engineer - InMoment, USA — Mar 2015 - April 2016
Software Engineer - McAfee, USA — Oct 2014 - Mar 2015
Asst. System Engineer/Role: Software Engineer - Tata Consultancy Services, INDIA — Sept 2010 - July 2012

Publications/Abstracts/Posters/Pre-print/Github

[27] **An Approach for Cross Platform Workload for Autonomous Database Systems**
Vishal Sharma

World Academy of Science, Engineering and Technology (WASET), 2026

[26] **A Framework to Unify Lost Knowledge**
Vishal Sharma
World Academy of Science, Engineering and Technology (WASET), 2026

[25] **The Metadata Impedance Mismatch between Databases and Programming Languages**
Vishal Sharma, Curtis Dyreson
34th *Conference on Information and Knowledge Management*, *CIKM*, 2025

[24] **Reproducibility Report for ACM SIGMOD 2024 Paper: "Temporal JSON Keyword Search"**
Chenhao Ma, Junchang Wang, Curtis Dyreson, Amani Shatwani, Sourav S Bhowmick, Vishal Sharma
50th *ACM International Conference on Management of Data*, Reproducibility Reports of the 2024 International Conference on Management of Data, 2024

[23] **Temporal JSON Keyword Search**
Curtis Dyreson, Amani Shatwani, Sourav S Bhowmick, Vishal Sharma
⋆ *Awards: Artifact Received, Artifact Evaluated and Results Reproduced*
50th *ACM International Conference on Management of Data*, *SIGMOD*, 2024

[22] **Artifact-Tolerant Clustering-Guided Contrastive Embedding Learning for Ophthalmic Images**
Min Shi, Anagha Lokhande, Mojtaba S Fazli, Vishal Sharma, Yu Tian, Yan Luo, Louis R Pasquale, Tobias Elze, Michael V Boland, Nazlee Zebardast, David S Friedman, Lucy Q Shen, Mengyu Wang
IEEE Journal of Biomedical and Health Informatics, 2023 (IF: 7.1)

[21] **Indexer++: Workload-Aware Online Index Tuning with Transformers and Reinforcement Learning**
Vishal Sharma, Curtis Dyreson
37th *ACM SIGAPP Symposium on Applied Computing*, *SAC*, 2022 (AR: 22%)

[20] **PyVisualFields: A Python Package for Visual Field Analysis**
Mohammad Eslami, Saber Kazeminasab, Vishal Sharma, Yangjiani Li, Mojtaba Fazli, Mengyu Wang, Nazlee Zebardast, and Tobias Elze
Translational Vision Science & Technology, TVST, ARVO, 2022 (IF: 3.28)

[19] **Mantis: Multiple Type and Attribute Index Selection using Deep Reinforcement Learning**
Vishal Sharma, Curtis Dyreson, Nicholas Flann
25th *ACM International Database Engineering & Applications Symposium*, IDEAS 2021 (AR: 28%)

[18] **Popularity vs Quality: Analyzing and Predicting the Success of Highly Rated Crowdfunded Projects on Amazon**
Vishal Sharma, Kyumin Lee, Curtis Dyreson
Springer Nature Computing, 2021 (IF: 3.7)

[17] **Automating and Analyzing Whole-Farm Carbon Models**
Aditi Maheshwari, Curtis Dyreson, Jennifer Reeve, Vishal Sharma, Anthony Whaley
7th *IEEE International Conference on Data Science and Analytics*, DSAA 2020 (AR: 27%)

[16] **Covid-19 Screening Using Residual Attention Network an Artificial Intelligence Approach**
Vishal Sharma, Curtis Dyreson
19th *IEEE International Conference on Machine Learning and Applications*, ICMLA 2020 (AR: 25%)

[15] **LinkSocial: Linking User Profiles Across Multiple Social Media Platforms**
Vishal Sharma, Curtis Dyreson
8th *IEEE International Conference on Big Knowledge, ICBK (in conjunction with ICDM)* 2018 (AR: 27%)

[14] **Predicting Highly Rated Crowdfunded Products**
Vishal Sharma, Kyumin Lee
10th *IEEE/ACM Advances in Social Networks Analysis and Mining*, ASONAM 2018 (AR: 16%)

[13] **Recommending Prime Spots of a Destination and Time to Visit from Geo-tagged Social Data**
Vishal Sharma, Kyumin Lee, Jinwook Chung
10th *IEEE International Conference on Collaborative Computing: Networking, Applications and Worksharing*, CollaborateCom 2014 (AR: 28%)

[12] **Supporting data aspects in pig latin**
Curtis Dyreson, Omar U. Florez, Akshay Thakre, Vishal Sharma
12th *ACM Aspect-oriented Software Development*, AOSD 2013 (AR: 25%)

[11] **The Multiverse Programming Paradigm: Programming with Values Annotated with Metadata**,
Vishal Sharma,
Graduate Research Symposium. Paper 87. https://digitalcommons.usu.edu/grs/87, (2014).

[10] **Role of generative artificial intelligence (AI) in generating course design and curriculum in dental public health**
Jyoti Mago, Vishal Sharma, Manoj Sharma
American Public Health Association, APHA, 2024

[9] **Role of generative artificial intelligence (AI) in oral health**
Jyoti Mago, Vishal Sharma, Manoj Sharma
American Public Health Association, APHA, 2024

[8] **Entity and Relation Identification in Unstructured Endoscopy Reports**
Nicholas Denson, Judith W Dexheimer, Julia Smith, Ryan Murphy, Vishal Sharma, Liu Xiaoxuan, Surya Prasath, Yizhao Ni, James Reigle, and Jasbir Dhaliwal
⋆ *Poster of Distinction*, *Digestive Disease Week*, DDW, 2023(IF : 3.49)

[7] **A Deep Autoencoder Model to Denoise Visual Fields in Glaucoma**
Vishal Sharma, Lucy Q Shen, Louis Pasquale, Tobias Elze, Michael V Boland, Sarah R Wellik, Gustavo De Moraes, Jonathan S Myers, Siamak Yousefi, Mengyu Wang
⋆ *Association for Research in Vision and Ophthalmology (ARVO) foundation travel grant, Denver, CO, 2022 (IF: 2.39)*

[6] **A Python Collection of Tools for Analyzing Visual Fields**
Saber Kazeminasab, Mohammad Eslami, Yangjiani Li, Mojtaba Fazli, Vishal Sharma, Mengyu Wang, Nazlee Zebardast, Tobias Elze
Association for Research in Vision and Ophthalmology, ARVO 2022 (IF: 2.39)

[5] **Evaluation of Deep Learning Visual Field Prediction Models for Clinical Relevance**
Mohammad Eslami, Miao Zhang, Julia Kim, Dolly Chang, Yangjiani Li, Saber Kazeminasab, Mojtaba Fazli, Vishal Sharma, Michael Boland, Nazlee Zebardast, Mengyu Wang, Tobias Elze
Association for Research in Vision and Ophthalmology, ARVO 2022 (IF: 2.39)

[4] **Glaucomatous Progressive Retinal Nerve Fiber Layer Thinning and Its Association With Patient Race**
Qingying Jin, Omar Halawa, Yangjiani Li, Mohammad Eslami, Saber Kazeminasab, Mojtaba Fazli, Vishal Sharma, Nazlee Zebardast, Mengyu Wang, Tobias Elze
Association for Research in Vision and Ophthalmology, ARVO 2022 (IF: 2.39)

[3] **The Impact of Race on the Relationship Between Cup-To-Disc Ratio and Glaucomatous VF Los**
Pingping Zhao, Yangjiani Li, Mohammad Eslami, Saber Kazeminasab, Mojtaba Fazli, Vishal Sharma, Omar Halawa, Nazlee Zebardast, Mengyu Wang, Tobias Elze
Association for Research in Vision and Ophthalmology, ARVO 2022 (IF: 2.39)

[2] **Speaker Diarization: Using Recurrent Neural Networks**
Vishal Sharma, Zekun Zhang, Zachary Neubert, Curtis Dyreson
arXiv:2006.05596, preprint, Github 2020

[1] **Multi Class Audio Classification Using Multi Layer Perceptron and Convolution Neural Network**
Vishal Sharma
https://doi.org/10.5281/zenodo.3988690, Github , 2020

Invited Research Presentations

2025	ACM International Conference on Information and Knowledge Management, CIKM, 2025, Seoul, Korea
2022	ACM SIGAPP Symposium On Applied Computing (SAC), Czech Republic
2021	ACM International Database Engineering & Applications Symposium (IDEAS), Montreal
2020	IEEE International Conference on Machine Learning and Applications (ICMLA), Miami
2018	IEEE International Conference on Big Knowledge (ICBK/ICDM), Singapore
2018	IEEE International Conference on Advances in Social Networks Analysis & Mining (ASONAM), Barcelona

Tools

Programming	Python, Java, C/C++, R
Data Preprocessing	OpenCV, NLTK, HuggingFace, spaCy, NetworkX
Data Visualization	Matplotlib, Plotly, ggplot2, seaborn, Orange, bigvis, Graphviz
Machine Learning	Scikit-learn, Weka
Deep Learning	TensorFlow, tflearn, Keras, Pytorch
Reinforcement Learning	OpenAI Gym, Ray RLlib, Keras-rl
Web Crawling	Scrapy, Selenium, Beautiful Soup
Databases	MySQL, PostgreSQL, MongoDB
Big Data	Elasticsearch, Kibana, Apache Pig, Apache Hadoop, Apache Spark, Apache Airflow
Open-source Contribution	tflearn, facebook/nevergrad, facebook/dime, aikorea/awesome-rl
Misc	LaTeX, Shell, Git, Inkscape

Patent

Methods to Denoise Visual Fields in Glaucoma
Inventors: Mengyu Wang, Vishal Sharma, Tobiaz Elze
(expired) U.S. Provisional Patent Application 63/335,125 (2022)

Academic Service & Leadership

2025	Reviewer	Int'l Journal of Software Engineering and Knowledge Engineering
2025	Technical PC	11th International Conference on Human and Social Analytics
2024	Technical PC	10th International Conference on Human and Social Analytics
2023	Technical PC	9th International Conference on Human and Social Analytics
2022	Reviewer	Review of Hypermedia and Multimedia (NRHM)
2022	Technical PC	8th International Conference on Human and Social Analytics
2021	Program Committee	IEEE BIBM Artificial Intelligence Techniques for BioMedicine and Health
2021	Technical PC	7th International Conference on Human and Social Analytics
2020	Program Committee	IJCAI Artificial Intelligence in Affective Computing (AffComp)
2020	Program Committee	IEEE BIBM Artificial Intelligence Techniques for BioMedicine and Health
2020	Program Committee	IEEE BIBM Artificial Intelligence & Big Data vs Pandemics
2020	Technical PC	6th International Conference on Human and Social Analytics
2019	Technical PC	5th International Conference on Human and Social Analytics
2018	Session Chair	ACM/IEEE ASONAM
2014	Student Volunteer	ACM SIGMOD

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